Revealing hidden jump strategies beyond conventional force-plate metrics

Short title: Curve shape beyond jump metrics

Abstract

Force-plate software summarises each countermovement jump (CMJ) with a few vendor metrics, and athletes with similar values are treated as equivalent. We tested that assumption with 805 routine CMJ trials from 212 male footballers in India. Principal component analysis of body-weight-normalised take-off and landing force–time curves retained four components (74.8% of variance), concentrated in the landing phase. Trials matched on jump height or relative peak force varied nearly as much as the full sample; vendor metrics, age and body mass explained 2.7–13.1% of each component. Most of this variation arose within athletes, so landing information needs averaging over trials and sessions.

Keywords: Countermovement jump; Principal component analysis; Force–time curve; Force plates; Athlete monitoring; Landing; Sports analytics; Football

Extended summary

High-performance centres invest in force plates and test athletes routinely, yet decisions about those athletes usually rest on a small number of vendor-computed values, such as jump height and peak force. Athletes with similar values are then treated as though they had jumped in the same way. We examine whether that inference is warranted and how much of the recorded information those values leave unused.

The analysis used 805 countermovement jumps from 212 male football players tested during routine monitoring at a private sports physiotherapy and performance centre in India. Each force–time curve was normalised to body weight; the take-off and landing phases were time-normalised separately and concatenated, and principal component analysis identified the main modes of variation. The dispersion of curve shapes among jumps matched on jump height, or on relative peak force, was compared with that of the whole sample, and the share of each mode explained by four vendor metrics, age and body mass was estimated.

Four components explained 74.8% of curve variance and lay mainly in the landing phase; the first described how rapidly downward momentum was arrested. Jumps matched on jump height were only marginally more similar in shape than the sample as a whole, and matching on relative peak force did not reduce dispersion to a detectable degree. The conventional metrics, age and body mass explained 2.7–13.1% of each component (0.3–11.8% out of sample). Between-athlete differences accounted for 10.0–29.6% of component variance, so most of the variation arose within athletes.

For managers of sports-science programmes, the landing is already recorded in every jump, at no further testing cost. Two simple landing descriptors accounted for about 56.8% of the leading landing component and could be added to routine reports, provided that decisions rest on several trials rather than on a single landing.

1 Introduction

Force plates are now standard equipment in many high-performance sports programmes, and the countermovement jump (CMJ) is the test most often performed on them. The test takes a few seconds, causes little fatigue and responds to training and to neuromuscular fatigue (Claudino et al., 2017; Gathercole et al., 2015). The vendor software identifies the phases of each jump and reports many variables per trial, such as jump height, peak forces, impulses and phase durations (McMahon et al., 2018). Few practitioners use all of them. Guidance recommends picking a handful that suit the purpose of testing, whether performance profiling, fatigue monitoring or rehabilitation (Bishop et al., 2023). We call these vendor-computed values vendor metrics. When two athletes post similar vendor metrics, their jumps are usually treated as equivalent.

That practice rests on a premise that is rarely tested. Many organisations reduce a rich measurement record to a few summary indicators, and for any of them the question is whether those indicators keep the information their decisions need. Motor-control research gives reason for doubt. A movement goal can be reached in more than one way, and trial-to-trial variation in execution is often functional rather than error (Bartlett et al., 2007; Davids et al., 2003; Preatoni et al., 2013). Two jumps of the same height may therefore follow different force–time paths, on the way up and on the way down.

We test that possibility in routine force-plate records of male football players in India, representing each jump by its full take-off and landing force–time curve. The analysis is organised around three questions. (1) What are the main modes of variation of these curves, and what do they mean mechanically? (2) Do trials with matched jump height, or matched relative concentric peak force, have more similar curves than the sample as a whole? (3) What share of each mode do four vendor metrics, age and body mass explain once repeated trials per athlete are allowed for? Two further analyses split the curve variation into between-athlete, between-session and within-session parts, and repeat the work with a second definition of the landing phase.

The methods are standard. What is new is the use made of them. Matching trials on a metric shows directly how much curve shape that metric leaves open. The landing, which every trial records but most CMJ studies set aside, is analysed together with take-off. And explained variance is estimated with the athlete, not the trial, as the independent unit. Section 2 reviews

prior work, Section 3 sets out the data and methods, and Section 4 reports the results. Sections 5 to 7 discuss them, draw out implications for practice and conclude.

2 Literature review

The three questions draw on two bodies of work, one built on discrete CMJ variables and the other on whole waveforms. Within the first, no single kinetic or temporal variable has been found to account for vertical-jump performance on its own (Dowling & Vamos, 1993). Better and poorer jumpers distribute force differently across unweighting, braking and propulsion (Sole et al., 2018), and force–time profiles differ by sex and sport (Laffaye et al., 2014). How force develops within each sub-phase is linked to the height achieved (Harry et al., 2018). More recent work separates outcome measures, such as jump height, from variables that describe how the jump was executed, such as countermovement depth and phase timing (Harry et al., 2024). The second kind of variable can reveal fatigue that jump height misses (Gathercole et al., 2015). Curve-level analyses of force, velocity and power have also shown training effects spread across the movement rather than at isolated points (Cormie et al., 2009).

Waveform methods avoid choosing variables in advance. Principal component analysis (PCA) of time-normalised curves, and functional PCA, express each curve as a mean plus a weighted sum of a few uncorrelated shape patterns, with one weight (score) per trial and pattern (Jolliffe, 2002; Jolliffe & Cadima, 2016; Ramsay & Silverman, 2005). Biomechanists have used these tools for gait in knee osteoarthritis (Deluzio & Astephen, 2007), knee motion during jumping (Ryan et al., 2006), running in Achilles tendinopathy (Donoghue et al., 2008) and rowing (Warmenhoven et al., 2019). Brandon et al. (2013) set out ways to read the resulting patterns. In jump research, waveform methods described performance-related features better than discrete points (Richter et al., 2014b), detected gains in jump performance (Floría et al., 2019), and extracted more information once curves were aligned on movement events (Moudy et al., 2018; Richter et al., 2014a).

This literature leaves two gaps. First, almost all of it stops at take-off. The same plates record the landing in the same trial, yet it is usually set aside, although there are reasons to look at it.

Leppänen et al. (2017) reported a higher risk of anterior cruciate ligament injury among young female basketball and floorball players who landed stiffly with high vertical force. Jump and landing measures show some value for injury prediction (Pedley et al., 2020). CMJ landings also change with the effort an athlete believes they are giving (Hite et al., 2025). Second, waveform studies show that curves contain more than discrete variables do, but they seldom measure how much curve shape the routinely reported metrics miss. We are not aware of a study that asks whether jumps matched on such a metric also have matched curves. Large routine datasets from youth football, collected under field conditions, make the question answerable.

3 Data and methodology

The dataset used to answer these questions was collected for monitoring, not for research, and several of the choices below follow from that. We describe its source and protocol first, then the construction of the curves and finally the statistical analysis.

3.1 Setting, participants and protocol

The design was retrospective and cross-sectional. The data were CMJ trials recorded during routine testing of male football players at a private sports physiotherapy and performance centre in India between November 2022 and April 2026. The centre shared the data with the authors under a signed non-disclosure and data-sharing agreement. All CMJ trials of male football players held in the centre's account were downloaded through the vendor's application programming interface: 833 trials from 215 athletes in 333 sessions. A session is one test occasion, at which an athlete performed one or more trials. Athlete, session and trial identifiers were replaced by pseudonymous codes before analysis. The analysis files hold no names, team names, birth dates, test dates or vendor identifiers. The first author keeps the code key separately. Age at test, to 0.1 year, was derived before the identifiers were removed. At their first analysed test, 164 of the 212 analysed athletes (77%) were younger than 18 years. **The study was approved by [Ethics committee name, approval number and date to be inserted]. [Written informed consent was obtained from adult athletes, and written consent from a parent or guardian, together with**

the athlete's assent, was obtained for athletes under 18 years.] / [The committee waived the requirement for consent for the secondary use of de-identified routine monitoring data.] [Delete the option that does not apply.] The study conformed to the Declaration of Helsinki.

Trials were recorded on VALD ForceDecks dual force plates (VALD Performance, Brisbane, Australia). Each plate measures the vertical ground reaction force under one foot, and the two signals were summed. The sampling rate was 1,000 Hz for most trials and 500 Hz for 59 of the 805 analysed trials (60 of the 833 extracted). Because every phase was later resampled to the same grid, this difference is unlikely to matter. Sessions contained a median of 2 trials, and athletes attended a median of 1 session (maximum 6). Centre staff ran the tests as part of routine assessment, not under a research protocol. Warm-up, verbal instructions and rest intervals were therefore not recorded with the data, and no landing technique was prescribed. For each trial the vendor software supplied the four vendor metrics used here: jump height (from flight time), concentric peak force, eccentric braking impulse and concentric impulse in the first 100 ms of the concentric phase. Contraction time, flight time, countermovement depth and peak landing force were also exported, for interpretation only.

3.2 Quality control and curve construction

Three events came from the vendor output. Movement onset t_{on} was taken directly. Take-off was $t_{\mathrm{to}} = t_{\mathrm{on}} + T_{\mathrm{c}}$, with T_{c} the contraction time, and touch-down was $t_{\mathrm{td}} = t_{\mathrm{to}} + T_{\mathrm{f}}$, with T_{f} the flight time (Harry et al., 2020; McMahon et al., 2018; Owen et al., 2014). The take-off phase spans t_{on} to t_{to}. Two landing definitions were used, because the literature offers no agreed end point for the CMJ landing. In the main analysis, the landing ends at the lowest point of the centre of mass (COM). At that instant the downward velocity gained during the fall has been reduced to zero, so the impact has been absorbed. COM velocity after touch-down was obtained from the impulse–momentum relationship (Linthorne, 2001; Street et al., 2001):

$$v(t) = -\frac{g\,T_{\mathrm{f}}}{2} + \int_{t_{\mathrm{td}}}^{t} \frac{F(s) - mg}{m}\,\mathrm{d}s, \qquad t_{\mathrm{lc}} = \min\{\, t > t_{\mathrm{td}} : v(t) \geq 0 \,\}. \tag{1}$$

Here F is total vertical force, m is the vendor body mass and $g = 9.81\,\mathrm{m\,s^{-2}}$. The starting value assumes equal take-off and touch-down speeds. The integral was evaluated as a cumulative sum over samples. The main landing phase is $[t_{\mathrm{td}}, t_{\mathrm{lc}}]$. In the sensitivity analysis, the landing phase is a fixed window from t_{td} to $t_{\mathrm{td}} + 1$ s, which is where the vendor's trial record ends.

Five exclusion rules were applied in sequence (Table 1). R1 removed trials whose vendor body mass was missing or not positive, since all curves are divided by body weight. R2 compared that mass with the mass implied by mean force during quiet standing, from the start of the record to 30 ms before movement onset. Trials that differed by more than 5% were removed, because they point to a wrongly entered mass or an athlete who was moving. R3 removed trials whose mean force between 20 ms after take-off and 20 ms before touch-down exceeded 50 N in absolute value, which indicates plate contact in flight or a mis-detected event. R4 removed trials with a missing vendor metric or missing age. R5, applied in the main analysis only, removed trials in which t_{lc} was not reached before the record ended. The rules took the sample from 833 trials (215 athletes) to 805 trials (212 athletes). The 1-s window, which does not need R5, kept 811 trials.

Table 1. Quality-control flow (main analysis).

Step	Trials	Sessions	Athletes
All male football CMJ trials	833	333	215
R1: valid body mass	828	329	212
R2: quiet-stance mass within 5% of body mass	813	328	212
R3: no force on plate during flight	811	327	212
R4: complete metrics and age	811	327	212
R5: end of landing (lowest COM) found	805	327	212

Note. Rules R1–R4 apply to both landing definitions; R5 applies only to the main landing definition (touch-down to the lowest point of the centre of mass), and the 1-s landing window therefore retained 811 trials. Sessions are test occasions on which an athlete performed one or more trials. COM = centre of mass.

Force was divided by body weight, $\tilde{F}(t) = F(t)/(mg)$. A value of 1 then means quiet standing, and athletes of different mass share one scale. Each phase $[t_a, t_b]$ was resampled by

linear interpolation at 101 equally spaced times,

$$\tilde{F}_k^{(p)} = \tilde{F}\Big(t_a + \tfrac{k}{100}\,(t_b - t_a)\Big), \qquad k = 0, 1, \ldots, 100. \tag{2}$$

Resampling each phase separately places take-off at the last point of the first block and touch-down at the first point of the second. This is a simple landmark registration (Moudy et al., 2018; Ramsay & Silverman, 2005; Richter et al., 2014a). A single time scale for the whole trial would put the same event at different percentages in different trials and mix timing with shape. Flight was left out because force is zero in the air and the only information in that segment is its length, which already defines flight-time jump height (Linthorne, 2001). Each trial therefore became a 202-element vector $\mathbf{x}_i$: 101 take-off values followed by 101 landing values.

3.3 Statistical analysis

PCA was run on the $n \times 202$ matrix of curves (Jolliffe, 2002; Jolliffe & Cadima, 2016). The curves were centred on the mean curve but not scaled, because every element is in the same unit (multiples of body weight). The decomposition is

$$\mathbf{S} = \frac{1}{n-1}\sum_{i=1}^{n}(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^{\top} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{\top}, \qquad z_{ik} = \mathbf{v}_k^{\top}(\mathbf{x}_i - \bar{\mathbf{x}}), \tag{3}$$

where $\mathbf{v}_k$ is the unit-length loading vector of component k, λ_k its eigenvalue and z_{ik} the score of trial i. Component k explains $\lambda_k / \sum_j \lambda_j$ of the total variance. The number of components was set at four before the analysis and was the same for both landing definitions. A fixed number avoids choosing the cut-off after seeing how components relate to the vendor metrics. The scree plot (Fig. 1B) was inspected only to describe the result. Signs were set so that the largest absolute loading of each vector was positive. Three tools aided interpretation. First, curves were reconstructed as $\bar{\mathbf{x}} \pm 2\sqrt{\lambda_k}\,\mathbf{v}_k$ (Brandon et al., 2013; Warmenhoven et al., 2019). Second, the landing share of each component was computed as $\sum_{j \in \mathcal{L}} v_{jk}^2$, with $\mathcal{L}$ the 101 landing elements. Third, scores were correlated (Pearson) with peak landing force relative to body weight, landing duration $t_{\mathrm{lc}} - t_{\mathrm{td}}$, contraction time, countermovement depth, the four vendor metrics, age and body mass.

The second question was addressed with matched bands. The first band contained trials whose jump height lay between its 45th and 55th percentiles. The second band used the same percentiles of concentric peak force relative to body weight. For each component, the dispersion ratio $\rho_k = \mathrm{SD}(z_k \mid i \in \mathcal{B})/\mathrm{SD}(z_k)$ divides the score SD inside band $\mathcal{B}$ by the SD in all trials. A ratio near 1 means that matching on the metric leaves curve shape as variable as before. A ratio well below 1 means that the metric restricts curve shape. For the curve as a whole, the ratio was $\rho = \{\mathrm{tr}(\mathbf{S}_\mathcal{B})/\mathrm{tr}(\mathbf{S})\}^{1/2}$, with $\mathbf{S}_\mathcal{B}$ the covariance matrix inside the band. Confidence intervals (CIs, 95%) for ρ_k were percentile intervals from 2,000 bootstrap samples (Efron & Tibshirani, 1993). The resampling unit was the athlete: each bootstrap sample drew athletes with replacement and kept all their trials. Band limits and PCA loadings were held fixed, so the intervals are conditional on the estimated components.

The third question was addressed by regression. Each score was regressed by ordinary least squares on the four vendor metrics, age and body mass: $z_{ik} = \beta_{0k} + \sum_{q=1}^{6} \beta_{qk}\, w_{iq} + \varepsilon_{ik}$. The coefficient of determination R^2_k is the share of component k that these six variables explain. In-sample R^2 overstates fit, and trials from one athlete are correlated. Three checks were therefore added. In five-fold cross-validation, athletes, not trials, were assigned at random to folds (Roberts et al., 2017). Out-of-sample R^2 was $1 - \sum_i (z_{ik} - \hat{z}_{ik}^{(-f)})^2 / \sum_i (z_{ik} - \bar{z}_k)^2$, where $\hat{z}_{ik}^{(-f)}$ comes from the model fitted without the fold that holds trial i. The model was also refitted on one randomly chosen trial per athlete, and on athlete means of all variables.

To see whether scores describe stable athlete characteristics, each score was split into variance components with a three-level random-intercept model, $z_{ajt} = \mu + a_a + b_{aj} + e_{ajt}$. Here $a_a \sim N(0, \sigma^2_\mathrm{A})$, $b_{aj} \sim N(0, \sigma^2_\mathrm{S})$ and $e_{ajt} \sim N(0, \sigma^2_\mathrm{T})$ are the athlete, session-within-athlete and trial-within-session effects. The model was fitted by restricted maximum likelihood (Pinheiro & Bates, 2000). The athlete share $\sigma^2_\mathrm{A}/(\sigma^2_\mathrm{A} + \sigma^2_\mathrm{S} + \sigma^2_\mathrm{T})$ is the single-trial intraclass correlation (Hopkins, 2000; Koo & Li, 2016). Only 83 of the 212 athletes had more than one session, so the session share is imprecise. CIs for all three shares came from 2,000 athlete-level bootstrap samples, with athletes relabelled within each sample so that repeated draws counted as separate athletes. The model was refitted with age at test as a fixed effect.

All steps were rerun with the 1-s landing window. The jump-height band was also widened or

narrowed (40th–60th percentile; median ± 1 cm; median ± 2 cm). Results are given as estimates with 95% CIs and read descriptively. No significance tests were run, so no multiplicity correction applies. One R script (R 4.5.3; R Core Team, 2026) performed every step, using dplyr 1.2.1, tidyr 1.3.2, ggplot2 4.0.3 (Wickham, 2016), patchwork 1.3.2 and nlme 3.1.170 (Pinheiro & Bates, 2000). PCA used the base function `prcomp`. All random steps (fold assignment, the one-trial draw and both bootstraps) used seed 2026. The script writes every table, figure and number in this paper. A rerun from the de-identified data reproduced all 42 values checked against an earlier, independent run of the same analysis. Reporting followed Hopkins et al. (2009).

4 Results

The results follow the order of the three questions. The sample and the modes of variation come first, followed by the matched bands, the regressions, the variance components and the sensitivity analysis.

4.1 Sample and modes of variation

The main analysis included 805 trials from 212 athletes in 327 sessions (Table 1). The 1-s window kept 811 trials, since R5 removed 6 trials only from the main analysis. Median age at first analysed test was 14.1 years (interquartile range 12.0–16.8; range 8.8–37.9). Athletes contributed 3.8 trials on average (maximum 16). Mean jump height was 28.2 ± 8.4 cm. Peak landing force spanned 2.19 to 14.88 body weights (Table 2). One trial had zero countermovement depth and zero braking impulse. It passed all five rules and was kept.

Table 2. Characteristics of the analysed sample (805 trials, 212 athletes).

Variable	*n*	Mean ± SD	Median (IQR)	Range
Age at first test (years)	212	16.01 ± 5.94	14.10 (12.00–16.80)	8.80–37.90
Body mass at first test (kg)	212	53.31 ± 15.90	51.73 (40.86–64.03)	25.03–95.97
Jump height, flight time (cm)	805	28.19 ± 8.43	26.96 (21.62–34.69)	8.54–51.47
Concentric peak force (N)	805	1269.26 ± 420.74	1227.01 (923.47–1582.99)	473.52–3073.22
Concentric peak force (× BW)	805	2.35 ± 0.38	2.29 (2.11–2.51)	1.65–4.70
Eccentric braking impulse (N s)	805	27.80 ± 15.43	24.60 (16.59–35.70)	0.00–95.18
Concentric impulse, first 100 ms (N s)	805	62.24 ± 26.66	58.70 (41.76–79.14)	1.51–187.93
Peak landing force (× BW)	805	5.39 ± 1.97	4.97 (3.88–6.67)	2.19–14.88
Contraction time (s)	805	0.78 ± 0.19	0.77 (0.66–0.90)	0.31–1.48
Countermovement depth (cm)	805	−26.55 ± 9.90	−27.13 (−33.55–−20.71)	−52.39–0.00
Touch-down–lowest COM (s)	805	0.25 ± 0.11	0.24 (0.15–0.31)	0.07–0.93
Trials per athlete	212	3.80 ± 2.28	3.00 (2.00–5.00)	1.00–16.00
Trials per session	327	2.46 ± 0.78	2.00 (2.00–3.00)	1.00–7.00
Sessions per athlete	212	1.54 ± 0.80	1.00 (1.00–2.00)	1.00–6.00

Note. Age and body mass are per athlete at the first analysed trial; all other rows are per trial unless stated. Jump height is derived from flight time. Countermovement depth is the vendor estimate of centre-of-mass displacement (negative downwards); one trial with zero depth and zero braking impulse satisfied all quality-control rules and was retained. Touch-down to lowest COM is the duration of the landing phase in the main analysis. BW = body weight; COM = centre of mass; IQR = interquartile range.

The mean curve (Fig. 1A) showed the familiar take-off sequence of unweighting, braking and propulsion, followed by a landing with one large force peak whose height varied widely between trials. Landing was far more variable than take-off. The landing block held 92.9% of the summed point-wise variance. PC1 explained 37.3% of the variance, and PC1 to PC4 together explained 74.8% (Fig. 1B). Ten components were needed to reach 93.8%.

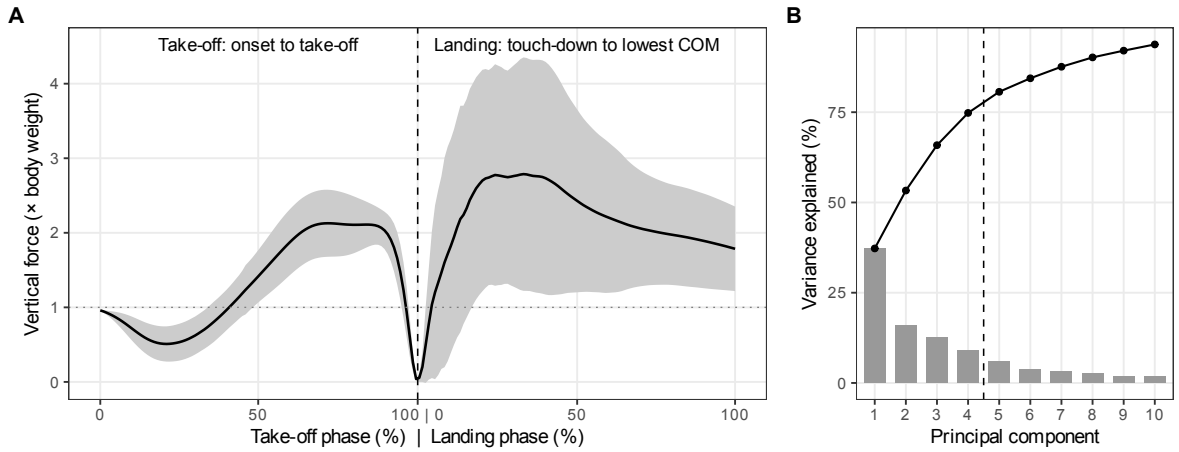


Fig. 1. (A) Mean body-weight-normalised vertical force (line) ± 1 SD (shaded) of the 805 analysed countermovement jumps. The take-off phase (movement onset to take-off) and the landing phase (touch-down to the lowest point of the centre of mass, COM) were each time-normalised to 0–100% and concatenated; flight is omitted. The dotted line marks body weight. (B) Variance explained by each of the first ten principal components (bars) and cumulatively (line); the dashed line marks the four retained components.

Between 99.3% and 99.9% of each retained component's squared loadings fell in the landing block (Table 3). This concentration follows partly from the method. Unscaled covariance PCA weights each point by its variance, and landing points vary most. PC1 (37.3%) separated two landing types (Fig. 2). In one, force climbed to a high peak near 40% of the landing and stayed well above body weight until the COM reached its lowest point. In the other, force peaked early and then fell away. Landing duration was the strongest correlate of PC1 ($r = -0.75$). A high score therefore marks a short, stiff landing that stops the downward motion quickly. PC1 correlated moderately with peak landing force ($r = 0.39$) but was almost unrelated to contraction time ($r = -0.04$) and countermovement depth ($r = 0.01$). PC2, PC3 and PC4 (16.0%, 12.6% and 8.9%) captured when the landing peaks occurred and how many there were. They contrasted a single early peak, a single later peak and two peaks. Their largest loadings sat at 26%, 16% and 9% of the landing. Every component correlated with peak landing force ($r = 0.22$–0.45). No correlation with contraction time exceeded 0.15 in absolute value, none with countermovement depth exceeded 0.18, and none with a vendor metric exceeded 0.34.

Table 3. Retained principal components of the concatenated take-off and landing curve (main analysis).

Component	Variance (%)	Cumulative (%)	Landing share of loading (%)	Largest loading at	Pearson r with LPF	JH	CT	CMD	LD
PC1	37.3	37.3	99.7	Landing 40%	0.39	0.13	−0.04	0.01	−0.75
PC2	16.0	53.3	99.9	Landing 26%	0.22	0.13	0.02	0.00	−0.14
PC3	12.6	65.9	99.5	Landing 16%	0.39	0.34	0.00	−0.14	−0.32
PC4	8.9	74.8	99.3	Landing 9%	0.45	0.24	0.15	−0.18	−0.04

Note. Landing share of loading is the percentage of the squared loading vector in the landing block (the remainder is in the take-off block). Largest loading at gives the phase and the percentage of the phase at which the loading has its largest absolute value. LPF = peak landing force (× BW); JH = jump height; CT = contraction time; CMD = countermovement depth; LD = landing duration (touch-down to lowest COM). Signs are fixed so that the largest absolute loading is positive.

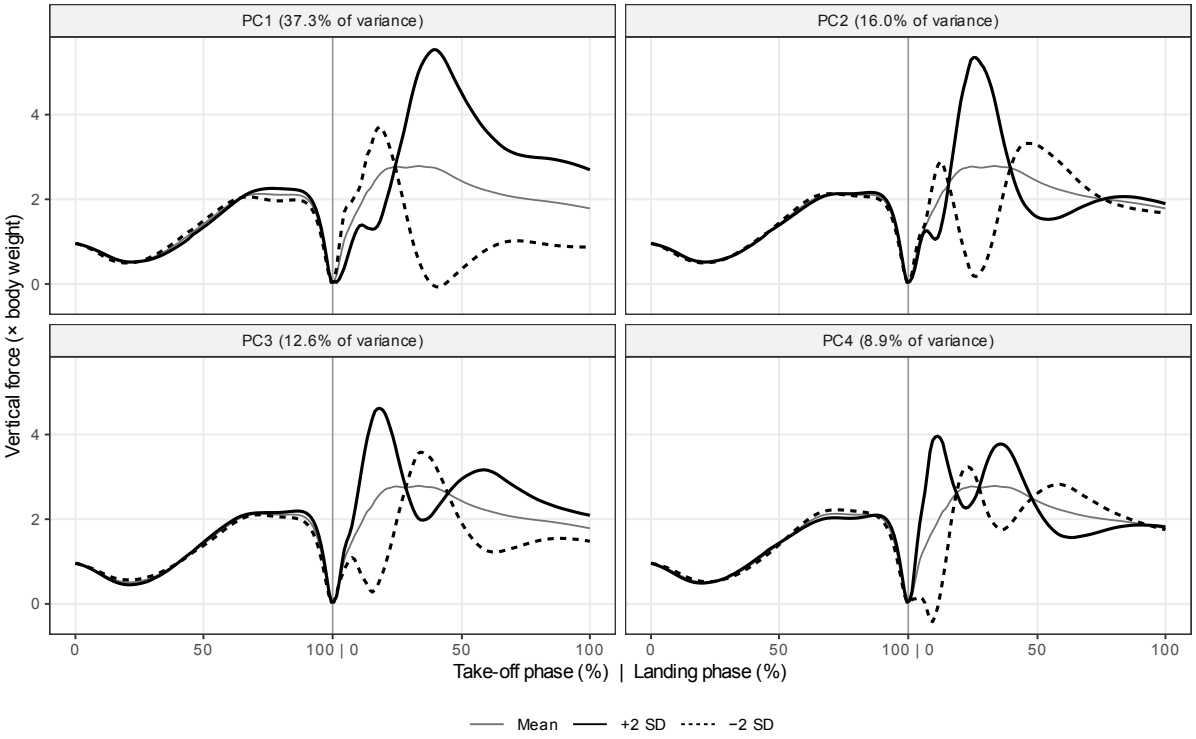


Fig. 2. Interpretation of the four retained principal components (main analysis). Each panel shows the mean curve (thin grey line) and the mean curve plus (solid black line) or minus (dashed black line) two standard deviations of the component score. Values below zero in the reconstructions are an artefact of adding a linear component to the mean and do not correspond to observed trials. The vertical line separates the take-off and landing phases.

4.2 Matching, explained variance and sensitivity

Once the components had been described, we asked how far conventional metrics pin them down. The jump-height band held 82 trials from 51 athletes. Their jump height was 27.1 ± 0.65 cm, an SD about 13 times smaller than in the full sample. Take-off curves in the band looked alike, but landing curves did not (Fig. 3A). For PC1, matching reduced the SD by a small amount (ratio 0.89; 95% CI 0.70–1.06). Ratios for PC2 to PC4 lay between 0.95–1.17, and all four CIs contained 1 (Table 4; Fig. 3B). The whole-curve ratio was 1.01. Matching on relative concentric peak force produced no detectable narrowing either: component ratios were 1.00–1.05, and the whole-curve ratio was 1.00. Changing the band width moved the PC1 ratio only within 0.87–0.89. Across all components and widths the ratios lay between 0.84–1.17.

Table 4. Dispersion of component scores in the full sample and among trials matched on jump height or on relative concentric peak force (main analysis).

		Jump-height band		Relative peak-force band	
Component	SD, full sample	SD	Ratio (95% CI)	SD	Ratio (95% CI)
PC1	7.28	6.45	0.89 (0.70–1.06)	7.33	1.01 (0.77–1.16)
PC2	4.77	4.62	0.97 (0.77–1.16)	5.01	1.05 (0.82–1.24)
PC3	4.22	4.03	0.95 (0.74–1.13)	4.25	1.00 (0.81–1.17)
PC4	3.56	4.16	1.17 (0.81–1.44)	3.75	1.05 (0.87–1.23)
Whole curve[a]	11.92	11.99	1.01	11.95	1.00

Note. Matched bands comprise trials between the 45th and 55th percentiles of jump height (26.0–28.2 cm; 82 trials, 51 athletes) or of concentric peak force relative to body weight (2.25–2.32 BW; 81 trials, 58 athletes). Ratio = SD in the band divided by SD in the full sample; 95% CI from 2,000 athlete-level bootstrap resamples, conditional on the estimated components. Component SDs depend on the number of grid points and are comparable only within this table; ratios are scale-free. [a]Square root of the summed point-wise variance of the 202-point curve.

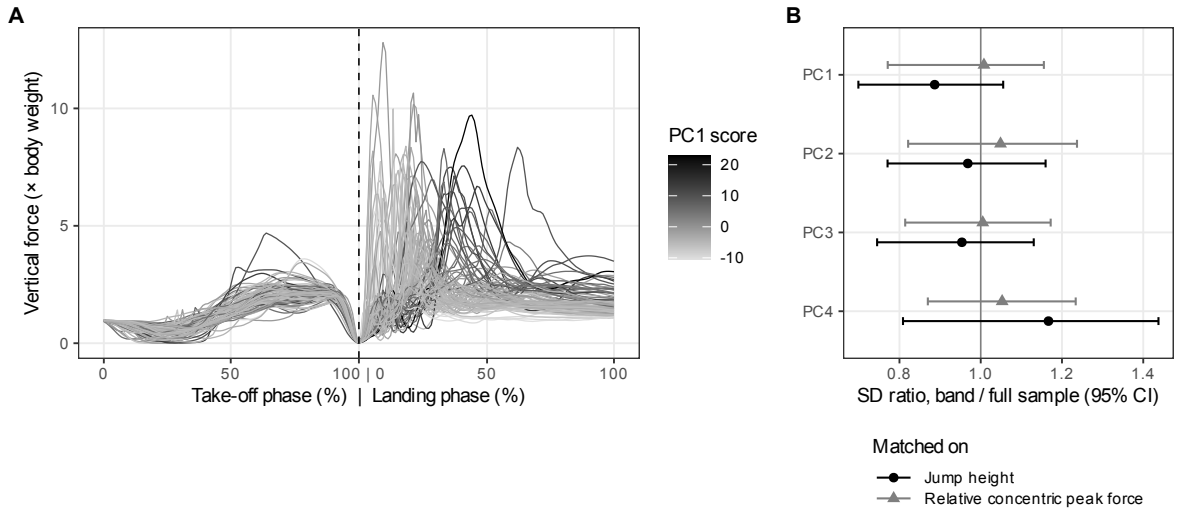


Fig. 3. Trials matched on jump height do not share a common force–time curve. (A) Curves of the 82 trials between the 45th and 55th percentiles of jump height (27.1 ± 0.65 cm), shaded from light grey (low) to black (high) by their PC1 score. (B) Ratio of the standard deviation of each component score among trials matched on jump height (black circles) or on relative concentric peak force (grey triangles) to that in the full sample, with athlete-level bootstrap 95% confidence intervals. A ratio of 1 indicates that matching on the metric leaves curve-shape variation unchanged.

The six predictors explained 2.7–13.1% of the variance of the four components in-sample, leaving 86.9–97.3% unexplained (Table 5; Fig. 4). Athlete-grouped cross-validation gave lower values (0.3–11.8%). Values were somewhat higher with one trial per athlete (4.3–17.8%) or with athlete means (5.4–21.2%), as expected when averaging removes trial-level noise, but they remained low.

Table 5. Variance of each component score explained by the four vendor metrics, age and body mass, under four evaluation schemes (main analysis, R^2 in %).

Component	In-sample	Adjusted	Athlete CV	One trial per athlete	Athlete means
PC1	7.7	7.0	4.0	10.0	11.1
PC2	2.7	2.0	0.3	4.3	5.4
PC3	13.1	12.4	11.8	17.8	21.2
PC4	9.1	8.4	6.3	11.1	13.8

Note. Predictors: jump height, concentric peak force, eccentric braking impulse, concentric impulse over the first 100 ms, age and body mass (ordinary least squares). In-sample and adjusted values use all 805 trials. Athlete CV = out-of-sample R^2 from five-fold cross-validation with athletes allocated to folds. One trial per athlete = one randomly drawn trial per athlete (212 trials). Athlete means = scores and predictors averaged within athlete.

Differences between athletes accounted for 10.0–29.6% of score variance (Table 6). The upper limits of the bootstrap CIs lay in the range 23–47%. Differences between sessions of the same athlete accounted for 2.3–34.9%, and differences between trials within a session for 35.5–74.8%. PC1 was the most athlete-specific component, with 29.6% of its variance between athletes (95% CI 13–47) and 34.9% between sessions. For PC2, PC3 and PC4, the within-session share was 61.6%, 74.8% and 69.6%. Adding age as a covariate changed the between-athlete shares by at most 7.1 percentage points (adjusted range 7.1–28.2%). Age at test therefore does not account for the low athlete shares.

Table 6. Decomposition of component-score variance between athletes, sessions and trials (main analysis, % of total variance).

Component	Between athletes	Between sessions	Within session	Between athletes, age-adjusted
PC1	29.6 (13–47)	34.9 (16–53)	35.5 (26–46)	28.2
PC2	21.4 (4–37)	17.0 (3–36)	61.6 (50–73)	20.6
PC3	22.9 (12–31)	2.3 (0–14)	74.8 (63–85)	15.9
PC4	10.0 (0–23)	20.3 (9–32)	69.6 (59–79)	7.1

Note. Three-level random-intercept model (athlete, session within athlete, trial within session) fitted by restricted maximum likelihood. Values in parentheses are 95% CIs from 2,000 athlete-level bootstrap resamples. Only 83 of the 212 athletes contributed more than one session, so the between-session share is imprecise. Age-adjusted = the same model with age at test as a fixed effect.

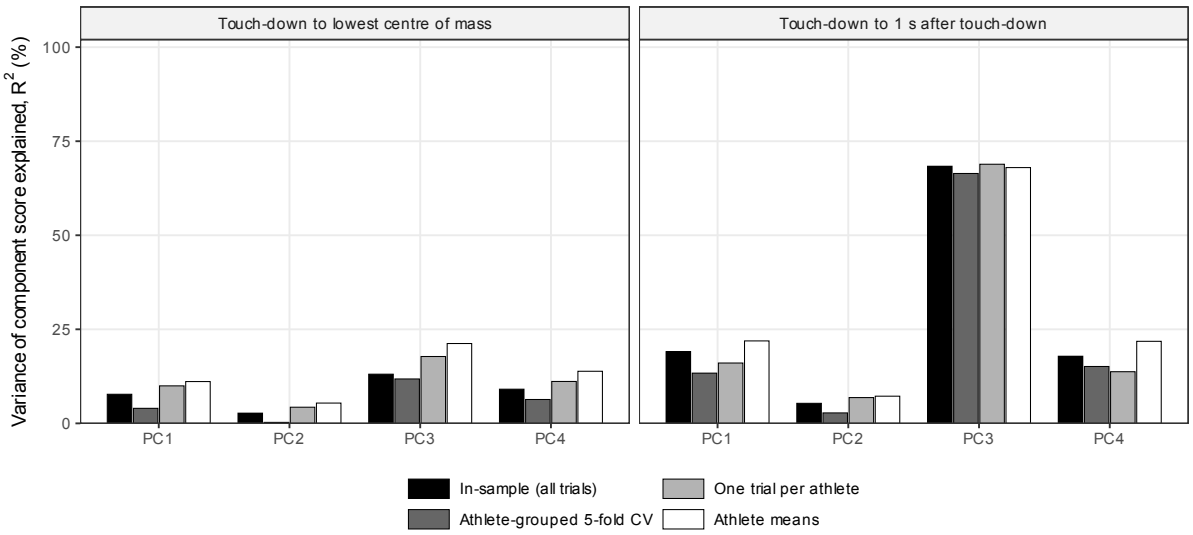


Fig. 4. Proportion of the variance of each component score explained by the four vendor metrics, age and body mass, under four evaluation schemes (bars from black to white: in-sample, athlete-grouped cross-validation, one trial per athlete, athlete means), for the main landing definition (left) and the 1-s landing window (right). CV, cross-validation with athletes allocated to folds.

The 1-s landing window changed the picture in one respect. With this window, impact occupied only the opening few per cent of the landing block, and the remainder was close to quiet standing (Fig. 5). The four components explained 62.2% of the variance. PC3 moved to the take-off block, which held 96.3% of its squared loading. It correlated strongly with contraction time ($r = -0.79$) and with relative concentric peak force ($r = 0.77$). The six predictors

explained most of this component (R^2 = 68.3%; 66.4% under cross-validation). The three landing components stayed poorly explained (R^2 = 5.3–19.1%; Table 7; Fig. 4). PC1 was again a landing component, linked to peak landing force (r = 0.71) and landing duration (r = −0.57). Jump-height matching narrowed PC1 a little more than in the main analysis (ratio 0.82; 95% CI 0.69–0.96), while the whole-curve ratio stayed at 1.01. Matching on relative concentric peak force narrowed the take-off component PC3 (ratio 0.74; 95% CI 0.57–0.87). The matched-band comparison can therefore detect a constraint when a metric does fix part of the curve.

Table 7. Main analysis compared with the sensitivity analysis in which the landing phase ended 1 s after touch-down.

Quantity	Main: landing to lowest COM	Sensitivity: 1-s landing
Trials / athletes after QC	805 / 212	811 / 212
Variance explained, PC1–PC4 (%)	37.3, 16.0, 12.6, 8.9	20.2, 16.6, 15.1, 10.2
Cumulative, PC1–PC4 (%)	74.8	62.2
Landing share of loading, PC1–PC4 (%)	99.7, 99.9, 99.5, 99.3	96.4, 92.5, 3.7, 80.2
PC1 SD ratio, jump-height band (95% CI)	0.89 (0.70–1.06)	0.82 (0.69–0.96)
PC3 SD ratio, relative-peak-force band (95% CI)	1.00 (0.81–1.17)	0.74 (0.57–0.87)
Whole-curve SD ratio, jump-height band	1.01	1.01
In-sample R^2, PC1–PC4 (%)	7.7, 2.7, 13.1, 9.1	19.1, 5.3, 68.3, 17.8
Athlete-CV R^2, PC1–PC4 (%)	4.0, 0.3, 11.8, 6.3	13.4, 2.8, 66.4, 15.1
Between-athlete variance, PC1–PC4 (%)	29.6, 21.4, 22.9, 10.0	45.3, 18.7, 26.5, 31.6
r(PC1, landing peak force)	0.39	0.71
r(PC3, contraction time)	0.00	−0.79

Note. COM = centre of mass; CI = confidence interval; CV = cross-validation. Between-athlete variance is taken from the three-level random-intercept model. In the 1-s window PC3 is a take-off component (Fig. 5), which explains its correlation with contraction time and its narrower dispersion among trials matched on relative peak force.

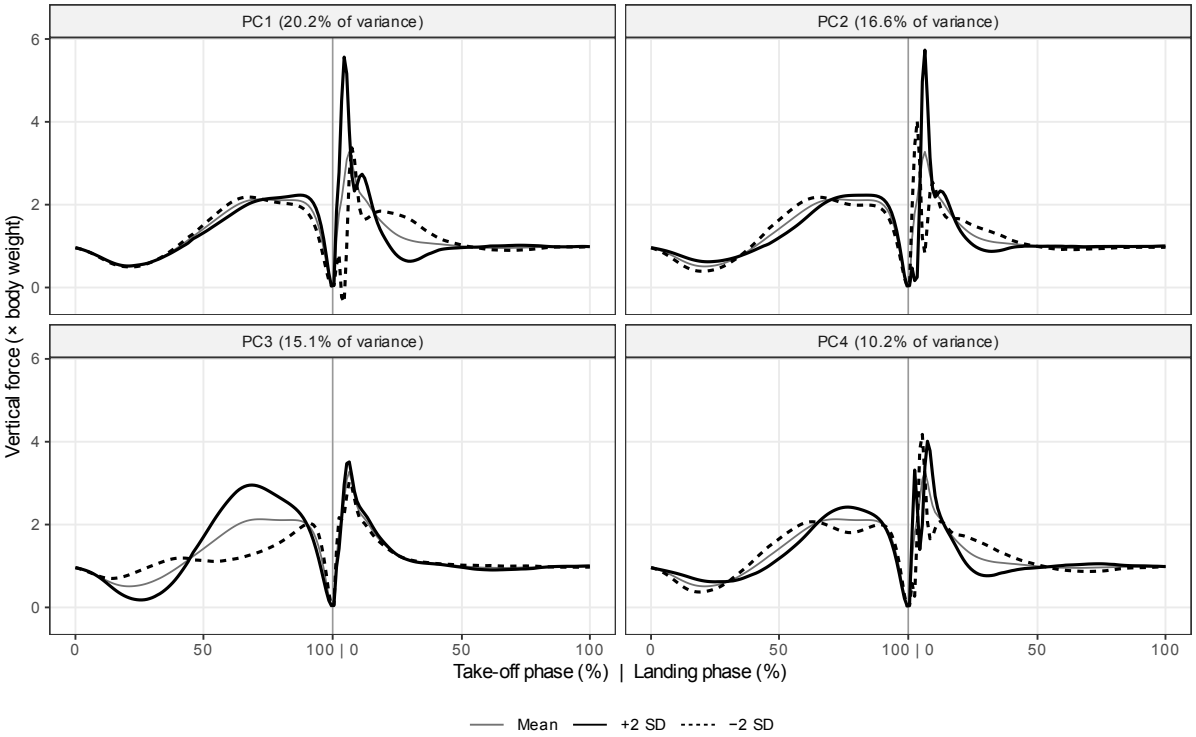


Fig. 5. Sensitivity analysis with the landing phase ending 1 s after touch-down: mean curve (thin grey line) and the mean curve plus (solid black line) or minus (dashed black line) two standard deviations of each retained component. Under this definition PC3 is located almost entirely in the take-off phase (96.3% of its squared loading) and describes the duration and amplitude of the propulsive force.

Two results carry forward from this section. The vendor metrics left most of the landing variation unexplained, and most of that variation lay within athletes, between sessions and between trials.

5 Discussion

Taken together, these results answer the question posed at the outset, whether CMJs with similar vendor metrics have similar force–time curves, within the limits of a retrospective design. On a common force scale, the main modes of variation lie in the landing, which held 92.9% of the point-wise variance. The leading mode describes how quickly the athlete's downward motion is stopped. Matching trials on jump height or relative peak force made their curves only slightly more alike, although the CIs still allow modest narrowing (for PC1 in the jump-height band the lower limit was 0.70). The four vendor metrics, age and body mass explained at most 13.1% of any landing component, and less out of sample. Equal vendor metrics, in short, do not imply

equal force–time curves. Nor are the components stable traits of athletes, since most of their variance arose within athletes.

These findings agree with earlier reports that discrete variables give an incomplete account of the jump. Dowling and Vamos (1993) found no single variable that explained jump performance well. Richter et al. (2014b) and Floría et al. (2019) showed that waveform methods capture performance-related features that discrete points miss. Those studies, like most CMJ waveform work (Cormie et al., 2009; Harry et al., 2018; Sole et al., 2018), analysed propulsion and related curve features to performance. We posed the reverse question: how much of the curve do the performance metrics account for? Matching on jump height gives a direct answer, because it holds the outcome nearly constant and shows what variation is left. The 1-s analysis locates that variation. There, a take-off component appeared, the metrics explained about two-thirds of it, and matching on relative peak force narrowed it. This fits the mechanical links among contraction time, peak force, impulse and jump height (Linthorne, 2001; McMahon et al., 2018), and the distinction between outcome and execution variables (Harry et al., 2024). The usual vendor metrics thus summarise take-off reasonably well and landing poorly. That is not a flaw in metrics designed to describe propulsion. What routine reports omit is the information in the landing, which every trial records.

PC1 can be read against the landing literature. Injury research has examined fast, high-force landings in young female athletes (Leppänen et al., 2017), and a review found that landing kinetics have some value as injury predictors (Pedley et al., 2020). PC1, which contrasts short, stiff landings with longer, softer ones, resembles this construct. The cited associations, however, come from young female athletes, and their relevance to male footballers is unknown. The link between PC1 and landing duration is mechanical. The main landing ends when downward velocity reaches zero, so the average force over the landing equals body weight plus the touch-down momentum divided by the landing time. A shorter landing must then carry higher forces.

The variance components limit how far the components can be called strategies. The between-athlete share was at most 29.6%, and no bootstrap upper limit exceeded 47%. By common benchmarks this means poor single-trial reliability (Koo & Li, 2016). Averaging helps. For PC1, the mean of 3 trials would estimate that session's value with a reliability of about 0.85,

but characterising the athlete would need several sessions. The landing differences between jumps of equal height therefore mostly reflect how a given trial or session was landed, not a fixed feature of the athlete. This is consistent with work treating movement variability as a normal part of skilled performance (Bartlett et al., 2007; Davids et al., 2003; Preatoni et al., 2013), and with the known sensitivity of CMJ variables to trial numbers and protocol (Cormack et al., 2008; Heishman et al., 2020). Landing also depends on instruction and perceived effort (Hite et al., 2025), neither of which was controlled. Here, *strategy* means the way a given trial was executed. Whether athletes hold stable landing strategies can only be judged from repeated landings under fixed instructions.

Two analytic choices shaped the results. The first is the landing end point. Ending at the lowest COM point confines the landing block to impact absorption and aligns landing timing across trials. A fixed 1-s window fills most of the block with quiet standing. It lowered the variance explained by four components from 74.8% to 62.2% and changed what the components described. The second is scaling. Unscaled covariance PCA weights each point by its variance. The take-off block held only 7.1% of the point-wise variance, so take-off contributes little to the leading components unless the landing is short or analysed on its own. Neither choice is wrong, but each answers a different question, and readers should interpret the matched-band and regression results with both in mind.

Because the data came from routine monitoring, they carry limitations that a laboratory study would avoid. Landing technique was not standardised, so some landing variation may reflect instruction, habit or effort rather than physical capacity. The landing end point depends on body mass and on a touch-down velocity inferred from flight time. Both carry error (Street et al., 2001), and the end point was not found in 6 trials. All athletes came from one private centre, to which they were referred or came by choice, so the sample does not represent Indian footballers in general. Ages ranged from 8.8 to 37.9 years, most athletes were adolescents, and maturity status, which affects jumping in young athletes (Lloyd & Oliver, 2012; Malina et al., 2015), was not recorded. Score dispersion was larger in adults than in athletes under 18 (PC1 SD 8.9 vs 6.5), so the pooled results lean towards the adolescent majority. The results may not apply to female athletes or other sports. The regression models were linear, with only four

vendor metrics as performance predictors. Adding landing variables would raise the explained share of the landing components. Linear time normalisation combines timing and amplitude differences in one analysis. The fifth and later components, which together held about a quarter of the variance, were left uninterpreted.

Several of these limits point to the next studies. Future work should test whether landing components are stable under repeated, standardised landings, relate them prospectively to injury, fatigue and return to play, include female athletes and other sports, and apply further registration, functional PCA (Ramsay & Silverman, 2005; Warmenhoven et al., 2019) or statistical parametric mapping (Pataky et al., 2013) to separate execution variability from consistent technique.

6 Managerial and practical implications

For coaches, sports scientists and the managers who run monitoring programmes, the findings lead to four practical points, each bounded by the fact that most of the variation lay within athletes. First, two jumps of equal height can be landed very differently. A report that ranks or groups athletes by jump height says little about how those jumps were absorbed. Because the differences arose mostly within athletes, they should not be read as fixed athlete characteristics. Second, every CMJ already records the landing at no extra testing cost. Two simple descriptors, landing duration to the lowest COM point and peak landing force relative to body weight, together explained 56.8% of PC1 but only 5.1–24.0% of PC2 to PC4. Both can be computed from the recorded signal without extra testing, which makes them candidates for routine reports. Third, landing shape varied substantially from trial to trial. Decisions should use the mean of several trials, or of repeated sessions, rather than one landing. Fourth, an unusual landing curve should prompt a closer look at how the athlete lands under fixed instructions. It is not a diagnosis, because no link to injury has been shown in footballers.

A broader point concerns any organisation that relies on instrumented measurement. Commercial software decides which summary numbers users see, and in these data those numbers described take-off well and landing poorly. A standard method applied to data already on file showed variation that the reports did not. Managers who buy measurement systems may benefit

from examining the full recorded signal before fixing the set of indicators they report. Extra variation helps decisions only if it repeats across trials, and here much of it did not.

7 Conclusion

We began by asking whether jumps that look alike in a force-plate report were performed alike. In this sample of male football players they were not: CMJs matched closely on jump height, or on relative peak force, differed in curve shape almost as much as unmatched jumps. The main differences lay in the landing, in particular in how quickly downward motion was stopped. Four vendor metrics, age and body mass explained little of this variation. Most of it arose within athletes, between sessions and between trials, so it should not be read as a stable individual strategy. Every CMJ trial already contains a landing, and it carries information that take-off metrics do not. Reports seldom use it. It deserves a trial in routine force-plate reporting, provided it is averaged over repeated trials and shown to be reproducible under standardised landing instructions.

Declarations

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and code availability

The R code that reproduces every reported result, and the de-identified trial-level data, are available in a public repository (link withheld for blind review; see title page). The raw force–time

recordings are held by the testing centre under a non-disclosure agreement and cannot be shared.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used AI tools to assist with language editing and code review. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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